

Code-layout, Readability and Reusability

Date	28 September 2026
Team Id	SWTID-2026-6259
Project Name	AI StudyBuddy API
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and Indentation are maintained throughout the code	Yes	Consistent Indentation and spacing are maintained
2	Proper File Structure	Files and folders are logically organized into separate modules.	Yes	Files and folders are logically organized
3	Meaningful variable Names	Variables clearly represent their purpose	Yes	Variable and function use meaningful names
4	Function/Met	Function are	Yes	Functions are

	Module Names	given clear and descriptive names based on their tasks		named according to their purpose
5	Code Comments	Comments are added to explain important code section	Yes	Comments are added where necessary
6	Modular Design	The backend is divided into reusable modules such as routes, controllers, middleware and services.	Yes	Authentication, AI, database and routes are separated into modules
7	No Redundant code	Duplicate and unnecessary code is avoided	Yes	Duplicate and unnecessary code is avoided
8	Error Handling	Errors and invalid request are handled with appropriate responses.	Yes	Errors are handled using proper validation and responses

Reusable Components/Modules:

S.No	Component/Module Name	Language/Technology	Where Reused	Reusability Level (High/Medium/Low)
------	-----------------------	---------------------	--------------	-------------------------------------

1	Authentication middleware	Node.js / Express/JWT	All protected APIs	High
2	Gemini AI Service	Node.js /Google Gemini	Summary, Flashcard, Quiz, Study plan	High
3	Database Connection module	Node.js /Mongoose	All database operations	High
4	File upload middleware	Node.js / Multer	Study material upload	Medium
5	API Route Modules	Express.js	Authentication, materials and AI features	High

Overall code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & structure	5	Project follows a well-organized modular structure
Readability	4	Meaningful names and clear module separation improve readability
Reusability	5	Authentication, AI and database modules can be reused
Documentation/Comments	4	Important modules and functions are documented
Overall Score	4.5	The project follows good coding practices and modular design

